

Preamble

Papers / Textbooks referenced:

- A versatile taxonomy of low-dimensional vortex models for unsteady aerodynamics (Darakananda & Eldredge)
- Low-Speed Aerodynamics (Katz & Plotkin)

1. Vortex lumping strategy maths

This section outlines a mathematical justification for the chosen vortex lumping strategy. Note: MAV-R is designed to be scalable, and more vortex lumping strategies are able to be added if need be.

1.1. Vorticity and circulation

First, let's run some first principles definitions.

The **vorticity** of the fluid is given by ζ (Katz & Plotkin), or ω (Darakananda & Eldredge), and is the measure of how much the fluid vector field ‘rotates’ about a point $\mathbf{x}$. That is to say, the vorticity is the **curl** of the fluid vector velocity field; if the velocity of the field is given as

$$\mathbf{q} = \begin{pmatrix} u \\ v \end{pmatrix} \quad (1.1)$$

, then the vorticity ω is given as

$$\omega = \nabla \times \mathbf{q} \quad (1.2)$$

Consider an arbitrary open surface S bounded in three-dimensional space by a closed curve C , as shown in Figure 1.

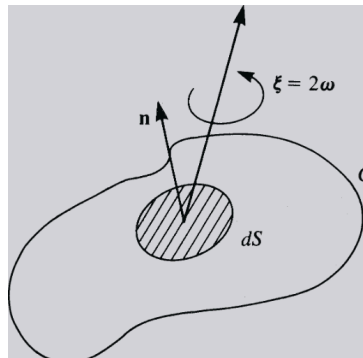


Figure 1: Closed surface in 3-D Space. Note here, the vorticity is denoted as ξ .

As it turns out, through Stokes' theorem (Katz & Plotkin), the vorticity ω on the surface S can be related to the line integral around C :

$$\int_S \nabla \times \mathbf{q} \cdot \mathbf{n} \, dS = \oint_C \mathbf{q} \, dl \quad (1.3)$$

, the RHS of which is the definition of circulation. Thus, if we consider the two-dimensional case, where $\mathbf{n} \, dS$ is simply dA (out of the page), then

$$\mathbf{\Gamma} = \oint_C \mathbf{q} \, dl = \int \nabla \times \mathbf{q} \, dA = \int \boldsymbol{\omega} \, dA \quad (1.4)$$

In 2-dimensional space, this circulation vector $\mathbf{\Gamma}$ is simply $\Gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. For the sake of simplicity / physical interpretation, we can think of just the magnitude; the directional vector will come into play when calculating cross products in the upcoming step. That is, for now, we consider:

$$\Gamma = \int \omega \, dA \quad (1.5)$$

Let us consider a differential area element dA , visualised as a tiny square of dimensions δx and δy . We can consider a (roughly) uniform vorticity distribution across such a small area, that is, $\delta\omega_x = \frac{\partial\omega}{\partial x}\delta x \approx 0, \delta\omega_y = \frac{\partial\omega}{\partial y}\delta y \approx 0$ in this dA element. Thus, a differential circulation contribution can be expressed as simply:

$$d\Gamma = \omega dA \quad (1.6)$$

1.2. Fluid impulse

Fluids have mass, and in a velocity field they are moving. Thus, they have momentum. We use a quantity called the **linear impulse** of the fluid, which can be thought of as the momentum per unit density.

From Darakananda & Eldredge, the fluid impulse $\mathbf{P}$ is given by:

$$\mathbf{P} = \int \mathbf{x} \times \boldsymbol{\omega} \, dA + \oint \mathbf{x} \times \boldsymbol{\gamma} \, dl + \oint \mathbf{x} \times (\mathbf{n} \times \mathbf{u}_B) \, dl \quad (1.7)$$

Where in order, each integral represents:

- The ambient fluid impulse, represented by vorticity in the fluid. Using the result of (1.6), this becomes:

$$\int \mathbf{x} \times \boldsymbol{\omega} \, dA = \int \mathbf{x} \, d\Gamma \quad (1.8)$$

- The impulse contribution of the bound vortex distribution on the airfoil, where $\boldsymbol{\gamma}$ represents this vortex distribution, and l is the direction along the surface contour.

- The impulse contribution from the fluid travelling along the surface. $\mathbf{n}$ represents a unit orthogonal vector to the surface at a specific point along the surface contour $\mathbf{x}$, and $\mathbf{u}_B$ is the local fluid vector at that point tangential to the surface.

Given our airfoil is going to be represented as infinitesimally thin, the third integral will evaluate to zero, leaving us with only integrals 1 and 2. That is, the impulse of the fluid consists of contributions from the ambient vorticity in the fluid (integral 1), as well as the vorticity contribution of the bound vortex distribution (here represented as a vortex sheet) on the airfoil.

1.3. Discrete vortices and impulse modelling

Let us now move to a more discrete representation to get closer to how our vortex panel solver might interpret these quantities.

The important distinction is to do with the representation of the ambient vorticity. Let us represent this ambient field (not the bound vorticity on the airfoil surface) using N_v discrete vortices. That is, the vorticity of the fluid field (not including that which is attached to the airfoil) is represented by discrete vortices, instead of a continuous vorticity distribution, whatever N_v may be.

To enforce the no-penetration boundary condition on the airfoil surface, the induced velocity from these ambient vortices must be cancelled out with the bound vortices.

Cleverly, we can represent the total bound vortex sheet (on the airfoil) as a **superposition** of $N_v + 1$ vortex sheets:

$$\gamma = \sum_i^{N_v} \gamma_i + \gamma_B \quad (1.9)$$

, where each γ_i is chosen such that it perfectly cancels out the induced normal velocity from ambient vortex i , such that the no-penetration condition is enforced. The final γ_B is chosen to counteract the motion of the airfoil in the fluid (ie. akin to being subject to a free-stream).

Applying the conversion from (1.8), and converting to the discretised representation discussed, we can represent the fluid impulse as:

$$\mathbf{P} = \sum_i^{N_v} \left(\mathbf{x} \times \Gamma_i + \oint \mathbf{x} \times \gamma_i d\mathbf{l} \right) + \oint \mathbf{x} \times \gamma_B d\mathbf{l} = \mathbf{P}_A + \mathbf{P}_B \quad (1.10)$$

Where $\mathbf{P}_A$ represents the impulse contribution from the terms dependent on the ambient vortices, and $\mathbf{P}_B$ is the impulse contribution from the bound terms only (independent of the ambient vortices).

Thus, we only really care about $\mathbf{P}_A$ because we are concerned with transferring / combining circulation of wake vortices, which are only ambient.

Thus we will investigate:

$$\mathbf{P}_A = \sum_i^{N_v} \left(\mathbf{x} \times \Gamma_i + \oint \mathbf{x} \times \gamma_i d\mathbf{l} \right) \quad (1.11)$$

If each γ_i exists to counteract the circulation Γ_i of vortex i , then by Kelvin's circulation conservation theorem, the total circulation around the airfoil for the bound sheet γ_i must be $-\Gamma_i$, that is, γ_i is proportional to Γ_i .

That is, each inner sum term is proportional to Γ_i and dependent on $\mathbf{x}_i$, and thus can be written as a function $\mathbf{p}(\mathbf{x}_i)$, that is,

$$\mathbf{P}_A = \sum_i^{N_v} \Gamma_i \mathbf{p}(\mathbf{x}_i) \quad (1.12)$$

Where:

$$\mathbf{p}(\mathbf{x}_i) = \mathbf{F}(\mathbf{x}_i) + \mathbf{B}(\mathbf{x}_i) \quad (1.13)$$

, where $\mathbf{F}(\mathbf{x}_i)$ is the free space ambient vortex impulse contribution per unit circulation, or:

$$\mathbf{F}(\mathbf{x}_i) = \mathbf{x}_i \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} y_i \\ -x_i \\ 0 \end{pmatrix} \quad (1.14)$$

, and $\mathbf{B}(\mathbf{x}_i)$ is the bound vortex impulse contribution per unit circulation of vortex i .

Applying the chain rule, we can express the rate of change of linear impulse of the fluid as:

$$\frac{d\mathbf{P}_A}{dt} = \sum_i^{N_v} \left(\Gamma_i \frac{d\mathbf{p}}{d\mathbf{x}_i} \cdot \dot{\mathbf{x}}_i + \dot{\Gamma}_i \mathbf{p}(\mathbf{x}_i) \right) \quad (1.15)$$

Darakananda & Eldredge mention that transferring / combining circulation will cause certain vortices to have a non-zero rate of change of circulation (due to one combining into the other). From (1.15), this means the second term in each summation is potentially non-zero.

This is a problem, as by lumping vortices together, if not done carefully, can cause a non-zero extra term in the rate of change of linear impulse of the fluid. That is, an unwanted 'spurious' force is created, since:

$$F \propto \frac{d\mathbf{P}_A}{dt} \quad (1.16)$$

Ideally, we don't want any extra spurious force to arise, and thus when vortex lumping we want to conserve impulse, that is:

$$\frac{d\mathbf{P}_A}{dt} = 0 \quad (1.17)$$

1.4. Far-field vortices and assumptions

We have seen that from (1.11), the impulse of the flow field which we are concerned with conserving depends on both ambient ‘free-space’ terms $\mathbf{F}(\mathbf{x}_i)$ (ie. vortices not bound to the airfoil), and bound vortex terms $\mathbf{B}(\mathbf{x}_i)$.

From both (1.11) and (1.13), we can see that $\mathbf{B}(\mathbf{x}_i)$ depends on γ_i , the required vortex circulation distribution to counteract the vortex i . Specifically, it counteracts the normal component of the induced velocity on the airfoil from vortex i . From potential flow theory, the induced velocity of a vortex j on a point i is given by:

$$\begin{pmatrix} u \\ w \end{pmatrix}_{ij} = \frac{\Gamma_j}{2\pi((z_i - z_j)^2 + (x_i - x_j)^2)} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{pmatrix} x_i - x_j \\ z_i - z_j \end{pmatrix} \quad (1.18)$$

from which we observe that the induced velocity decreases with $\approx \frac{1}{r^2}$. That is, faraway vortices will induce a low magnitude velocity on the airfoil, from which γ_i and γ_i ’s rate of change will be weak.

That is, for regions of space far away from the airfoil (where the origin is located), we can assume that the fluid field is approximately ‘free-space’, and neglect the bound vortex terms $\mathbf{B}(\mathbf{x}_i)$.

Thus:

$$\mathbf{p}(\mathbf{x}_i) \approx \mathbf{F}(\mathbf{x}_i) = \begin{pmatrix} y_i \\ -x_i \end{pmatrix} \quad (1.19)$$

1.5. Lumping vortex logic

Suppose we have a group j of same-signed far-field vortices with circulations Γ_j , which we will combine into a single ‘lumped’ vortex with circulation Γ_L .

Kelvin’s circulation theorem says that circulation has to be conserved, thus:

$$\Gamma_L = \sum_j \Gamma_j \quad (1.20)$$

The impulse of this group after the lumping is given by:

$$\mathbf{P}_{\text{new}} = \Gamma_L \mathbf{p}(\mathbf{x}_L) \quad (1.21)$$

, where $\mathbf{x}_L$ indicates the position of the new lumped vortex. From (1.19), this is then simplified as:

$$\mathbf{P}_{\text{new}} = \Gamma_L \begin{pmatrix} y_L \\ -x_L \end{pmatrix} \quad (1.22)$$

And

$$\mathbf{P}_{\text{old}} = \sum_j \Gamma_j \mathbf{p}(\mathbf{x}_j) = \sum_j \Gamma_j \begin{pmatrix} y_j \\ -x_j \end{pmatrix} \quad (1.23)$$

Since we need:

$$\mathbf{P}_{\text{new}} = \mathbf{P}_{\text{old}} \quad (1.24)$$

That yields the equation

$$\Gamma_L \begin{pmatrix} y_L \\ -x_L \end{pmatrix} = \sum_j \Gamma_j \begin{pmatrix} y_j \\ -x_j \end{pmatrix} \quad (1.25)$$

Which, we can split into the y and x components:

$$\Gamma_L y_L = \sum_j \Gamma_j y_j \quad (1.26)$$

$$\Gamma_L x_L = \sum_j \Gamma_j x_j \quad (1.27)$$

And thus recombining yields:

$$\Gamma_L \mathbf{x}_L = \sum_j \Gamma_j \mathbf{x}_j \quad (1.28)$$

That is:

$$\mathbf{x}_L = \frac{\sum_j \Gamma_j \mathbf{x}_j}{\Gamma_L} = \frac{\sum_j \Gamma_j \mathbf{x}_j}{\sum_j \Gamma_j} \quad (1.29)$$

In other words, by applying the far-field approximation assumption, we reach a lumping strategy in which neighbouring, similar-signed vortices are combined using the following rules:

$$\Gamma_L = \sum_j \Gamma_j \quad (1.30)$$

$$\mathbf{x}_L = \frac{\sum_j \Gamma_j \mathbf{x}_j}{\sum_j \Gamma_j} \quad (1.31)$$

Which is essentially combining their circulation, and making the new lumped vortex position being the weighted centroid of all previous j vortices.

1.6. Some things to note / potentially explore

- How far away defines ‘far-field’ (perhaps $|\mathbf{x}_i| > 10c$?)
- Merge only same signed vortices (avoids close to 0 floating-point errors and sensitivity)
- Max number of vortices that can be combined in a single ‘lumping’.
- If already-lumped vortices can be re-lumped or not (due to multiplicative errors)